Clinical Need and Technical Background for an Exhaled Carbon Dioxide and Hypoventilation Monitor

Summary

This review examines the clinical and technical evidence underpinning Project 06, a bench-only prototype that samples airway gas, confirms the presence of exhaled carbon dioxide (CO_2) and derives a breathing rate from the CO_2 signal. The clinical literature indicates that sustained exhaled CO_2 is the accepted means of excluding oesophageal intubation, and that capnography identifies respiratory depression during procedural sedation earlier than pulse oximetry. The technical literature indicates that clinical sidestream capnographs respond within fractions of a second, whereas the manufacturer of the low-cost non-dispersive infrared (NDIR) module selected for this project rates its response time at up to 120 s. These findings support the position stated in the project card: the prototype must measure and report its own response time and must not claim breath-by-breath capnography unless the measured data justify it.

1. Introduction

1.1 Purpose

The purpose of this review is to provide a traceable evidence base for the Clinical need and Background sections of the design dossier (D9) and to identify published values that may inform the engineering requirements (D1). The review is organised around the four clinical requirements (CR-1 to CR-4) defined in Project Card 06.

1.2 Method

Sources were identified through web and library database searches conducted on 26 October 2026. The following inclusion criteria were applied:

- peer-reviewed journal articles and professional society guidelines published in or after 2020;
- three exceptions, explicitly identified as such: the manufacturer's datasheet for the selected sensor, which is the only source of its rated performance; the Canadian Anesthesiologists' Society position paper on procedural sedation, first published in

2018 and reissued with modifications in 2026; and the international standard for respiratory gas monitors, ISO 80601-2-55:2018, which is the current edition;

- relevance to at least one clinical requirement or to the sensing approach described in the project card.

In-text citations are numbered in order of first appearance and presented as hyperlinks placed immediately before the full stop of the sentence they support. The numbered reference list at the end of the document follows the same order.

1.3 Scope

This document reports published evidence only. It contains no measurements of the prototype, and no statement in it should be read as evidence that the prototype meets a requirement. The device described in Project 06 is a bench-only prototype and is not intended for clinical use.

2. Clinical Need

2.1 Unrecognised oesophageal intubation (CR-1, CR-2)

Unrecognised oesophageal intubation remains a cause of preventable death. An international consensus guideline from the Project for Universal Management of Airways identifies sustained exhaled CO_2 as the principal test for excluding oesophageal placement of a tracheal tube [1] (https://doi.org/10.1111/anae.15817). The guideline defines sustained exhaled CO_2 by four criteria: a waveform that rises during exhalation and falls during inspiration, consistent or increasing amplitude over at least seven breaths, a peak more than 1 kPa (7.5 mmHg) above baseline, and a value that is clinically appropriate [1] (https://doi.org/10.1111/anae.15817).

The guideline further states that an exhaled CO_2 partial pressure below 1 kPa (7.5 mmHg) is highly unlikely following correct tracheal intubation in a patient with spontaneous circulation [1] (https://doi.org/10.1111/anae.15817). Where the criteria are not satisfied, the recommended default action is removal of the tube and ventilation by facemask or supraglottic airway [1] (https://doi.org/10.1111/anae.15817). The authors report that deaths continue to occur despite the widely promoted principle of “no trace = wrong place” [1] (https://doi.org/10.1111/anae.15817).

Relevance to the project. The seven-breath criterion provides a published reference for the "stated number of breaths" required by CR-1. The abrupt disappearance of a previously present CO_2 signal is the event that CR-2 requires the device to treat as critical.

2.2 Users of the information

Project Card 06 identifies the clinicians who act on the information the device produces. The proceduralist and the sedation nurse respond to an alarm by stopping drug administration, repositioning the airway and ventilating the patient. The anaesthetist called to a deteriorating sedation needs to know when the deterioration began. The intended change in practice is that an absent CO_2 trace is recognised immediately rather than after oxygen saturation falls, and that a blocked sampling line is distinguished from a patient who has stopped breathing.

Relevance to the project. These users require an unambiguous display (CR-1), an immediate critical alarm (CR-2), a technical fault that cannot be mistaken for apnoea (CR-3) and a recorded trend that shows when a fall began (CR-4).

2.3 Capnography during procedural sedation (CR-4)

Professional guidance in Canada requires CO_2 monitoring during sedation outside the operating theatre. The Canadian Anesthesiologists' Society states that continuous end-tidal CO_2 monitoring with waveform capnography is mandatory during general anaesthesia and during moderate or deep procedural sedation [2] (https://doi.org/10.1007/s12630-018-1230-z). The same position paper recommends capnography during minimal sedation to detect progression to a deeper level of sedation with associated respiratory depression [2] (https://doi.org/10.1007/s12630-018-1230-z). It also identifies capnography as particularly useful when ventilation cannot be observed directly, for example during magnetic resonance imaging [2] (https://doi.org/10.1007/s12630-018-1230-z).

Recent clinical studies support the premise that the CO_2 signal reveals respiratory depression before oxygen saturation declines. In a prospective study of 101 children undergoing procedural sedation, capnography detected respiratory depression a median of 35 s earlier than pulse oximetry [3] (https://doi.org/10.3390/children12070938). In the same cohort, capnography detected every apnoea episode and 76.9% of hypopnoeic hypoventilation episodes, and combined monitoring identified 93 events compared with 53 identified by pulse oximetry alone [3] (https://doi.org/10.3390/children12070938).

A comparable interval was observed in adults. In 85 patients sedated for flexible bronchoscopy, the median delay between the onset of apnoea on capnography and a decline in SpO_2 of at least 4% was 32 s [4] (https://doi.org/10.21037/apm-21-1009). Apnoea, defined as cessation of airflow for more than 10 s, was recorded in every patient in that study [4] (https://doi.org/10.21037/apm-21-1009).

Studies of larger populations associate capnography with improved safety outcomes. A service evaluation of 1,401 sedated procedures reported a reduction in adverse events from 22.1 to 12.7 per 100 procedures following the introduction of capnography, with an adjusted odds ratio of 0.57 (95% CI 0.42–0.77) [5] (https://doi.org/10.3389/fmed.2022.867536). A multicentre evaluation of 1,189 adults aged 65 years or older undergoing sedated gastrointestinal endoscopy found that a visible capnography display was associated with a shorter duration of subclinical hypoxaemia (incidence rate ratio 0.70, 95% CI 0.54–0.91), although the incidence of hypoxaemia did not differ between groups [6] (https://doi.org/10.2147/CIA.S637350).

Despite these recommendations, adoption remains incomplete. A European survey cited by Català Altarriba and colleagues found that only 46% of paediatric emergency departments use capnography routinely [3] (https://doi.org/10.3390/children12070938).

Strength of evidence. The clinical studies cited above are observational, retrospective or non-randomised, and the paediatric study was not powered to estimate sensitivity or specificity [3] (https://doi.org/10.3390/children12070938). None evaluated a low-cost NDIR module of the type used in this project, so their findings describe the value of capnography in general rather than the expected performance of the prototype.

2.4 Physiological reference values and definitions (CR-2, CR-4)

Normal end-tidal CO_2 is generally reported as 35–45 mmHg [7] (https://doi.org/10.7759/cureus.53289). Normal resting respiratory rate in adults ranges from 12 to 20 breaths/min [8] (https://doi.org/10.3390/s20216396). Respiratory rate monitoring is considered important for alerting clinicians when a patient receiving opioids is at risk of respiratory depression [8] (https://doi.org/10.3390/s20216396).

Sedation research distinguishes between two patterns of hypoventilation. Bradypnoeic (type 1) hypoventilation has been defined as a decreased respiratory rate with end-tidal CO_2 above 50 mmHg, whereas hypopnoeic (type 2) hypoventilation has been defined as end-tidal CO_2

below 30 mmHg, or a decline of more than 10 mmHg from baseline, with a normal or decreased respiratory rate [3] (https://doi.org/10.3390/children12070938). In that study, each condition had to persist for more than two consecutive readings, which were recorded every 10 s [3] (https://doi.org/10.3390/children12070938). Apnoea has been defined as the cessation of airflow for more than 10 s [4] (https://doi.org/10.21037/apm-21-1009).

Relevance to the project. These definitions show that a fall in respiratory rate and a fall in CO_2 are both recognised signs of hypoventilation, which supports CR-4. They are expressed in mmHg for clinical capnographs, so they cannot be applied directly to a module whose range ends at 5000 ppm; any threshold adopted for the prototype must be justified against the team's own calibration data.

3. Technical Background

3.1 Principle of non-dispersive infrared sensing

An NDIR sensor passes broadband infrared radiation through a gas cell and measures the intensity that reaches a detector. The attenuation caused by gas absorption follows the Beer–Lambert law, in which transmitted intensity decreases exponentially with gas concentration and optical path length [9] (https://doi.org/10.3390/mi15101203). A representative design comprises an infrared emitter, a gas cell and a thermopile detector fitted with an optical filter centred at 4.26 µm [9] (https://doi.org/10.3390/mi15101203). At this wavelength the absorption of water vapour is approximately three orders of magnitude weaker than that of CO_2, which gives the measurement its selectivity [9] (https://doi.org/10.3390/mi15101203).

3.2 Mainstream and sidestream sampling

Capnographs are classified by the location of the sensor. In mainstream systems the sensor is placed directly in the airway, whereas in sidestream systems gas is drawn through a sampling line to a sensor remote from the patient [7] (https://doi.org/10.7759/cureus.53289). Sidestream sampling introduces a time lag but can be applied to non-intubated patients [7] (https://doi.org/10.7759/cureus.53289). Changes in temperature and humidity along the sampling line may also affect measurement accuracy [7] (https://doi.org/10.7759/cureus.53289).

A study of 35 spontaneously breathing adults compared simultaneous mainstream and sidestream capnography at controlled rates of 10, 12 and 20 breaths/min [10] (https://doi.org/10.1038/s41598-024-75808-0). End-tidal CO_2 measured by the two techniques correlated strongly at every breathing rate ($r > 0.90$), yet the authors concluded that sidestream capnography cannot substitute for the more accurate mainstream technique [10] (https://doi.org/10.1038/s41598-024-75808-0).

Relevance to the project. The prototype uses a sidestream configuration consisting of a sampling line and water trap. The project card therefore requires the team to quantify how transport delay increases with line length.

3.3 Response time of clinical and low-cost sensors (CR-1)

The performance of clinical capnographs is governed by an international standard, ISO 80601-2-55, which specifies requirements for the basic safety and essential performance of respiratory gas monitors, including carbon dioxide monitoring [11] (https://www.iso.org/standard/67241.html). Only the abstract of this standard was consulted, so its specific rise-time and delay requirements are not reported here.

Rise time is defined as the time required for the measured CO_2 value to increase from 10% to 90% of its peak value [12] (https://doi.org/10.1007/s10877-025-01346-3). One commercial sidestream capnography system specifies a rise time of up to 190 ms with a 2 m sampling line [12] (https://doi.org/10.1007/s10877-025-01346-3).

The MH-Z19C module selected for this project is rated by its manufacturer at a response time (T90) of less than 120 s, an accuracy of ±(50 ppm + 5% of reading) and a preheat time of 1 min [13] (https://www.winsen-sensor.com/d/files/mh-z19c-pins%26terminal-type-co2-manual(ver1_2).pdf). The unit in use is labelled with a range of 400–5000 ppm (team records, docs/sensor-mhz19c.md), whereas exhaled gas contains approximately 5% CO_2 (50,000 ppm). Undiluted exhaled gas would therefore be expected to saturate the sensor.

The slow response is a property of this module rather than of NDIR sensing as such. A purpose-built compact NDIR sensor achieved a t90 of 3 s when switched between nitrogen and 10% CO_2 [9] (https://doi.org/10.3390/mi15101203). Low-cost NDIR sensors also differ considerably in accuracy and noise, and calibration against a reference analyser improved accuracy by approximately 65% in one evaluation [14] (https://doi.org/10.3390/s24175675).

Relevance to the project. A rated response time of up to 120 s, compared with a breath interval of a few seconds, indicates that the module is unlikely to resolve individual breaths. The manufacturer's figure is nonetheless a claim rather than a measurement, and T90 is not identical to the 10–90% rise time used for clinical systems. The team's own step-response and line-length measurements will determine whether the prototype supports breath-by-breath capnography or only a breath-averaged trend.

3.4 Sampling-line faults (CR-3)

A bench study of capnography sampling lines reported that loose, weakened or disconnected connections introduce measurement error through leakage and dilution of the gas sample [12] (https://doi.org/10.1007/s10877-025-01346-3). The same study assessed the volume of liquid each filter could tolerate before clogging, and found that several filters not matched to the monitor clogged earlier than expected [12] (https://doi.org/10.1007/s10877-025-01346-3). The study was funded by a capnography manufacturer and used simulated breathing only [12] (https://doi.org/10.1007/s10877-025-01346-3).

A 2024 clinical report describes how this type of fault can mimic a change in the patient. During general anaesthesia, a sidestream sampling line that had been stretched became partially loosened at the water trap, and end-tidal CO_2 fell from 32 to 24 mmHg while the patient's vital signs and ventilation were unchanged [15] (https://doi.org/10.4103/ija.ija_719_24). All values returned to the previous range immediately after the connection was tightened [15] (https://doi.org/10.4103/ija.ija_719_24). The author warns that failing to recognise a faulty sampling line may lead to an erroneous interpretation of the patient's condition, for example as low cardiac output [15] (https://doi.org/10.4103/ija.ija_719_24). This is a single-case report and represents a low level of evidence.

The manual for the selected module describes an automatic self-calibration performed every 24 hours after power-on [13] (https://www.winsen-sensor.com/d/files/mh-z19c-pins%26terminal-type-co2-manual(ver1_2).pdf). This feature assumes periodic exposure to clean air, a condition that may not hold when the sensor is connected to a sampling line carrying exhaled gas.

Relevance to the project. A disconnected line, which dilutes the sample with room air, and a blocked line, which prevents gas from reaching the sensor, may both produce a falling or absent CO_2 reading. This is the reason CR-3 requires such events to be reported as a technical

fault rather than as apnoea. The risk of liquid in the line supports the inclusion of a water trap, and the self-calibration behaviour is a candidate zero-drift failure mode for D8.

3.5 Precedent for bench evaluation of low-cost devices

The development of a low-cost portable ventilator illustrates a staged evaluation approach comparable to that required in Project 06. The prototype was first tested on a breathing simulator following a protocol drawn from ISO 80601-2-12:2020 and was then compared against a conventional anaesthesia ventilator while titrating to end-tidal CO_2 targets of 35–55 mmHg [16] (https://doi.org/10.1093/milmed/usae295). The authors acknowledge that simulator testing only approximates physiological conditions and that the device requires further validation before clinical use [16] (https://doi.org/10.1093/milmed/usae295).

A closer precedent is a handheld capnography device built from an infrared CO_2 sensor and an Arduino Mega 2560 microcontroller [17] (https://doi.org/10.1142/S1793545821500152). When compared with a reference, most differences fell within the 95% limits of agreement, with mean differences of 0.15 mmHg for end-tidal CO_2 and 0.40 breaths/min for respiratory rate [17] (https://doi.org/10.1142/S1793545821500152). The abstract does not state the type of infrared sensor or its response time [17] (https://doi.org/10.1142/S1793545821500152).

A 2025 technical report describes a battery-powered capnograph built around an ESP32 microcontroller and a SprintIR-6S NDIR sensor [18] (https://doi.org/10.7759/cureus.97324). The author selected this sensor for its fast response time and its ability to measure CO_2 concentrations of up to 20%, so that a waveform could be acquired during every breath [18] (https://doi.org/10.7759/cureus.97324). The design adds a differential pressure sensor in the sampling tubing to detect the respiratory phase, and a miniature pump that purges residual CO_2 from the sensing chamber between breaths [18] (https://doi.org/10.7759/cureus.97324). The author states that performance has not yet been tested against clinical-grade reference equipment and that long-term sensor drift has not been evaluated [18] (https://doi.org/10.7759/cureus.97324).

Relevance to the project. The ventilator study concerns a different device and is cited for its method only. The handheld capnograph shows that a microcontroller-based design can be compared against a reference instrument using limits of agreement, an analysis the team could apply in its validation sessions. The 2025 report shows the sensor characteristics a breath-by-breath design requires, namely a fast response and a range of up to 20% CO_2, which the MH-Z19C does not offer; its in-line pressure sensor also suggests one way to

detect flow independently of CO_2, which is relevant to CR-3. Together these studies support the use of a reference instrument, in this project the simulation laboratory capnograph, during calibration and validation sessions, and they model the cautious reporting expected of a bench prototype.

4. Implications for the Engineering Requirements

Table 1 relates each finding to the clinical requirement it informs. All values in the right-hand column are **proposed, to be justified or to be experimentally determined**; none has been verified on the prototype.

Table 1. Literature findings and proposed inputs to D1

CR	Finding from the literature	Proposed input to D1 (to be confirmed)
CR-1	Sustained CO_2 is assessed over at least seven breaths [1] (https://doi.org/10.1111/anae.15817)	Reference value for the "stated number of breaths", subject to the measured response time
CR-1	Clinical sidestream rise time is up to 190 ms at 2 m [12] (https://doi.org/10.1007/s10877-025-01346-3); MH-Z19C T90 is rated at less than 120 s [13] (https://www.winsen-sensor.com/d/files/mh-z19c-pins%26terminal-type-co2-manual(ver1_2).pdf)	A dedicated requirement to measure and report the 10–90% response time and its dependence on line length
CR-2	Apnoea is defined as cessation of airflow for more than 10 s [4] (https://doi.org/10.21037/apm-21-1009)	Candidate input for the latency of the critical alarm, limited by sensor speed

CR-3	Disconnection causes leakage and dilution; liquid can clog sampling lines [12] (https://doi.org/10.1007/s10877-025-01346-3); a loose line at the water trap lowered $EtCO_2$ from 32 to 24 mmHg with no change in the patient [15] (https://doi.org/10.4103/ija.ija_719_24)	Fault detection that does not depend on the CO_2 value alone, together with a water trap
CR-4	Bradypnoeic hypoventilation: low respiratory rate with $EtCO_2$ > 50 mmHg; hypopnoeic: $EtCO_2$ < 30 mmHg or a fall > 10 mmHg from baseline [3] (https://doi.org/10.3390/children12070938)	Advisory alarm on a sustained fall in derived rate or in CO_2 relative to the patient's own baseline; thresholds to be set from calibration data
CR-4	Normal adult respiratory rate is 12–20 breaths/min [8] (https://doi.org/10.3390/s20216396)	Reference range for the derived breathing rate
CR-4	CO_2 reveals respiratory depression a median of 32–35 s before SpO_2 [3] (https://doi.org/10.3390/children12070938), [4] (https://doi.org/10.21037/apm-21-1009)	The advisory alarm on a sustained fall should operate within a period that preserves this lead time

5. Limitations of this Review

- Findings were extracted from article pages and abstracts; the full texts should be read before any sentence is quoted in the final dossier.
- The clinical evidence is predominantly observational and was not obtained with low-cost sensors.
- Three sources fall outside the 2020 inclusion criterion and are identified as exceptions.

- The hypoventilation definitions are taken from a paediatric study and are expressed in mmHg for a clinical capnograph.
- Only the abstract of ISO 80601-2-55 was available, because the full standard is not openly accessible. Clinical response-time benchmarks were therefore taken from published literature. The prototype is a bench-only device and makes no claim of compliance with this standard.

6. Conclusion

The literature establishes a clear clinical need for reliable exhaled CO_2 monitoring, both to exclude oesophageal intubation and to detect hypoventilation during procedural sedation before oxygen saturation falls. It also shows that the value of such monitoring depends on a fast response and on the ability to distinguish a patient who has stopped breathing from a sampling line that has failed. Because the selected module is rated far slower than clinical capnographs, the most defensible position for the prototype is to measure its own response time, report it plainly and limit its claims to what those measurements support.

References

1. Chrimes N, Higgs A, Hagberg CA, et al. Preventing unrecognised oesophageal intubation: a consensus guideline from the Project for Universal Management of Airways and international airway societies. Anaesthesia. 2022;77:1395–1415. https://doi.org/10.1111/anae.15817
2. Dobson G, Chong MA, Chow L, et al. Procedural sedation: a position paper of the Canadian Anesthesiologists' Society. Canadian Journal of Anesthesia. 2018;65:1372–1384 (modified publication 2026). https://doi.org/10.1007/s12630-018-1230-z
3. Català Altarriba L, Yeh Hsi S, Ravit AM, Brió Sanagustín S, González-Rioja X. Non-invasive capnography versus pulse oximetry for early detection of respiratory depression during pediatric procedural sedation: a prospective observational study. Children. 2025;12:938. https://doi.org/10.3390/children12070938
4. Minami D, Murakami E, Shibata Y, et al. End-tidal capnographic monitoring during flexible bronchoscopy under fentanyl and midazolam sedation. Annals of Palliative Medicine. 2021;10(8):8665–8671. https://doi.org/10.21037/apm-21-1009

5. Corbett G, Pugh P, Herre J, et al. Service evaluation of the impact of capnography on the safety of procedural sedation. Frontiers in Medicine. 2022;9:867536. https://doi.org/10.3389/fmed.2022.867536
6. Wang P, Jiang S, Xu Y, et al. Continuous capnography and shorter duration of subclinical hypoxemia during sedated gastrointestinal endoscopy in older adults: a multicenter observer-blinded device evaluation. Clinical Interventions in Aging. 2026;21. https://doi.org/10.2147/CIA.S637350
7. Damam S, Meshram RJ, Taksande A, et al. Navigating pediatric capnography: a comprehensive review of scope and limitations. Cureus. 2024;16(1):e53289. https://doi.org/10.7759/cureus.53289
8. Nicolò A, Massaroni C, Schena E, Sacchetti M. The importance of respiratory rate monitoring: from healthcare to sport and exercise. Sensors. 2020;20(21):6396. https://doi.org/10.3390/s20216396
9. Xu M, Tian W, Lin Y, Xu Y, Tao J. Development of a compact NDIR CO_2 gas sensor for a portable gas analyzer. Micromachines. 2024;15(10):1203. https://doi.org/10.3390/mi15101203
10. Tolnai J, Rárosi F, Tóth I, Babik B, Novák Z, Peták F, Fodor GH. Relationships between capnogram parameters by mainstream and sidestream techniques at different breathing frequencies. Scientific Reports. 2024;14. https://doi.org/10.1038/s41598-024-75808-0
11. International Organization for Standardization. ISO 80601-2-55:2018 Medical electrical equipment — Part 2-55: Particular requirements for the basic safety and essential performance of respiratory gas monitors. Geneva: ISO; 2018 (abstract only; international standard). https://www.iso.org/standard/67241.html
12. Jones TL, Glozman D, Oglesby HJA, et al. Evaluation of accuracy, filter performance, and durability among capnography sampling lines: a bench study. Journal of Clinical Monitoring and Computing. 2025. https://doi.org/10.1007/s10877-025-01346-3
13. Winsen Electronics. MH-Z19C infrared CO_2 sensor user's manual, version 1.2. 2021 (manufacturer datasheet). https://www.winsen-sensor.com/d/files/mh-z19c-pins%26terminal-type-co2-manual(ver1_2).pdf
14. Dubey R, Telles A, Nikkel J, et al. Low-cost CO_2 NDIR sensors: performance evaluation and calibration using machine learning techniques. Sensors. 2024;24(17):5675. https://doi.org/10.3390/s24175675

15. Goneppanavar U. Troubleshooting the defect in the sampling line of the side stream capnograph. Indian Journal of Anaesthesia. 2024;68(10):921–922. https://doi.org/10.4103/ija.ija_719_24
16. Weeks MK, Widmann NJ, Nickel AJ, et al. Novel high-tech, low-cost ventilation for military use, mass casualty incidents, stockpiling, and underserved communities. Military Medicine. 2024;189(Suppl 3):823–831. https://doi.org/10.1093/milmed/usae295
17. Singh OP, El-Badawy IM, Malarvili MB. Design and validation of a handheld capnography device for cardiopulmonary assessment based on the Arduino platform. Journal of Innovative Optical Health Sciences. 2021;14(6):2150015. https://doi.org/10.1142/S1793545821500152
18. Rrmoku D. Design and development of a compact, portable nondispersive infrared (NDIR)-based capnography device for real-time end-tidal carbon dioxide (CO_2) monitoring. Cureus. 2025;17(11):e97324. https://doi.org/10.7759/cureus.97324